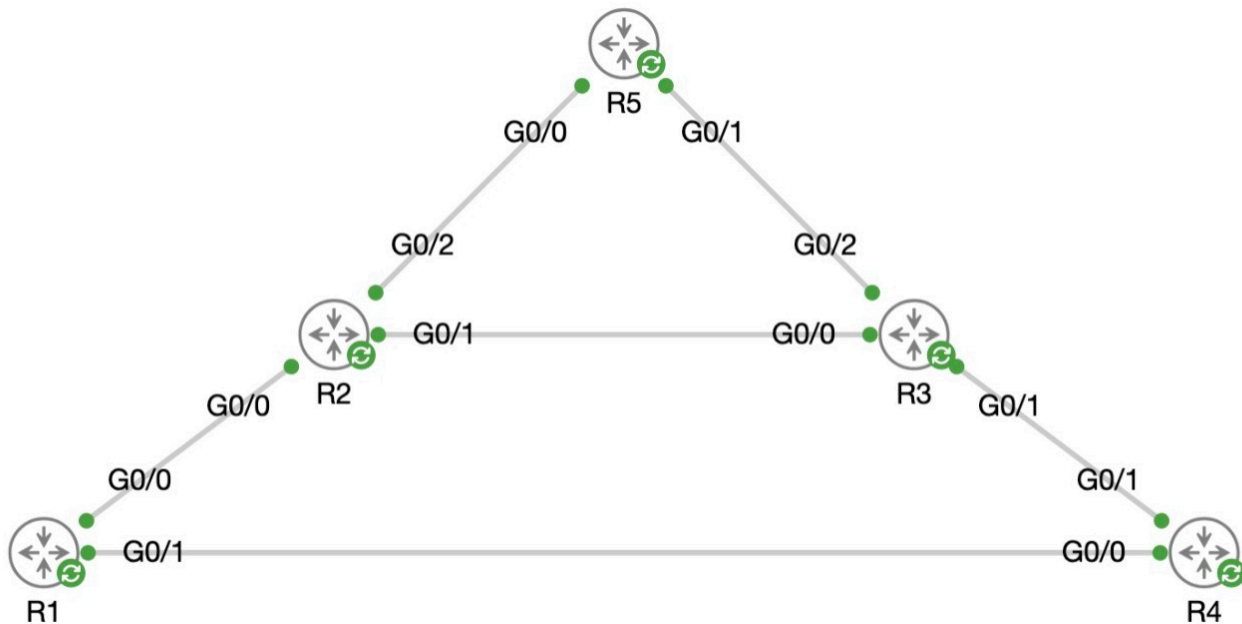


Network Design



IP Addressing Plan

R1 - R2 - 10.0.12.0/30

R1 - R4 - 10.0.14.0/30

R1 - R3 - 10.0.13.0/30

R4 - R5 - 10.0.45.0/30

Loopback0 for R1, R2, R3, R4, R5 - 1.1.1.1/32 to 5.5.5.5/32

Configuration

Interface configuration

R1:

```
en
conf t
hostname R1
int gi0/0
ip add 10.0.12.1 255.255.255.252
no sh
int gi0/1
ip add 10.0.14.1 255.255.255.252
no sh
int lo0
ip add 1.1.1.1 255.255.255.255
no sh
exit
```

R2:

```
en
conf t
hostname R2
int gi0/0
ip add 10.0.12.2 255.255.255.252
no sh
int gi0/1
ip add 10.0.23.1 255.255.255.252
no sh
int gi0/2
ip add 10.0.25.1 255.255.255.252
no sh
int lo0
ip add 2.2.2.2 255.255.255.255
no sh
```

```
exit
```

R3:

```
en
conf t
hostname R3
int gi0/0
ip add 10.0.23.2 255.255.255.252
no sh
int gi0/1
ip add 10.0.34.1 255.255.255.0
no sh
int gi0/2
ip add 10.0.35.1 255.255.255.252
no sh
int lo0
ip add 3.3.3.3 255.255.255.255
no sh
exit
```

R4:

```
en
conf t
hostname R4
int gi0/0
ip add 10.0.14.2 255.255.255.252
no sh
int gi0/1
ip add 10.0.34.2 255.255.255.252
no sh
int lo0
ip add 4.4.4.4 255.255.255.255
no sh
exit
```

R5:

```
en
conf t
hostname R5
int gi0/0
ip address 10.0.25.2 255.255.255.252
no sh
int gi0/1
ip add 10.0.35.2 255.255.255.252
no sh
int lo0
ip add 5.5.5.5 255.255.255.255
no sh
exit
```

Protocol Configuration

- On R1,

```
conf t

router eigrp ENTERPRISE
 address-family ipv4 unicast autonomous-system 100

  network 10.0.12.0 0.0.0.3
  network 10.0.14.0 0.0.0.3
  network 1.1.1.1 0.0.0.0

  af-interface GigabitEthernet0/0
    no passive-interface
  exit-af-interface

  af-interface GigabitEthernet0/1
    no passive-interface
  exit-af-interface

  af-interface Loopback0
    passive-interface
  exit-af-interface

exit-address-family
```

- On R2,

```
conf t

router eigrp ENTERPRISE
 address-family ipv4 unicast autonomous-system 100

    network 10.0.12.0 0.0.0.3
    network 10.0.23.0 0.0.0.3
    network 10.0.25.0 0.0.0.3
    network 10.2.0.0 0.0.0.255
    network 2.2.2.2 0.0.0.0

    af-interface GigabitEthernet0/0
      no passive-interface
    exit-af-interface

    af-interface GigabitEthernet0/1
      no passive-interface
    exit-af-interface

    af-interface GigabitEthernet0/2
      no passive-interface
    exit-af-interface

    af-interface Loopback0
      passive-interface
    exit-af-interface

exit-address-family
```

- On R3,

```
conf t

router eigrp ENTERPRISE
 address-family ipv4 unicast autonomous-system 100

    network 10.0.23.0 0.0.0.3
    network 10.0.34.0 0.0.0.3
    network 10.0.35.0 0.0.0.3
    network 10.3.0.0 0.0.0.255
    network 3.3.3.3 0.0.0.0
```

```

af-interface GigabitEthernet0/0
  no passive-interface
exit-af-interface

af-interface GigabitEthernet0/1
  no passive-interface
exit-af-interface

af-interface GigabitEthernet0/2
  no passive-interface
exit-af-interface

af-interface Loopback0
  passive-interface
exit-af-interface

exit-address-family

```

- On R4,

```

conf t

router eigrp ENTERPRISE
  address-family ipv4 unicast autonomous-system 100

    network 10.0.14.0 0.0.0.3
    network 10.0.34.0 0.0.0.3
    network 10.4.0.0 0.0.0.255
    network 4.4.4.4 0.0.0.0

  af-interface GigabitEthernet0/0
    no passive-interface
  exit-af-interface

  af-interface GigabitEthernet0/1
    no passive-interface
  exit-af-interface

  af-interface Loopback0
    passive-interface
  exit-af-interface

exit-address-family

```

- On R5,

```
conf t

router eigrp ENTERPRISE
  address-family ipv4 unicast autonomous-system 100

    network 10.0.25.0 0.0.0.3
    network 10.0.35.0 0.0.0.3
    network 10.5.0.0 0.0.0.255
    network 5.5.5.5 0.0.0.0

  af-interface GigabitEthernet0/0
    no passive-interface
  exit-af-interface

  af-interface GigabitEthernet0/1
    no passive-interface
  exit-af-interface

  af-interface Loopback0
    passive-interface
  exit-af-interface

exit-address-family
```